

Physics, 2018 EC, entrance exam, version 6

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes**Questions:** 54**Total marks:** 54

Name _____ Grade _____ ID No. _____ Date _____

1. In modern hospitals, MRI scanners are preferred over X-ray machines for imaging soft tissues such as the brain. (1)
This is because MRI technology is based on
- A. Nuclear magnetic properties of atoms B. High-frequency sound waves
C. Ionizing electromagnetic radiation D. Reflection of visible light
-
2. Ultrasound imaging is a safe, non-invasive technique commonly used to monitor fetal development during pregnancy. Unlike X-rays, it does not use ionizing radiation. This is because ultrasound technology is based on: (1)
- A. Nuclear magnetic resonance of hydrogen atoms
B. Reflection of high-frequency sound waves
C. Emission of gamma rays from radioactive tracers
D. Absorption of infrared radiation by tissues
-
3. Propose how principles of physics could be combined with biology to explain a complex biological process like nerve signal transmission. (1)
- A. By using fluid dynamics models to explain blood flow in neurons.
B. By using thermodynamics to calculate the energy released during signal transmission.
C. By applying quantum mechanics to explain neurotransmitter release.
D. By applying electrical circuit theory and ion movement to model action potentials.
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4. A radar station sends a pulse at speed c . The pulse hits a plane and returns to the station. The total time for the round trip is t . Which formula correctly calculates the distance (D) to the plane? (1)
- A. $D = ct$ B. $D = ct/2$
C. $D = ct^2$ D. $D = 2ct$
-
5. Radar systems used in aviation and defense can detect distant aircraft even during fog or at night. This is possible because radar systems operate using (1)
- A. Mechanical waves that require a medium
B. Electromagnetic waves that travel at the speed of light
C. Sound waves with very high intensity
D. Infrared radiation only
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6. Radiation therapy relies on high-energy particles to destroy cancer cells. This is an application of: (1)
- A. Nuclear Physics B. Fluid Mechanics
C. Classical Mechanics D. Acoustics
-
7. Which imaging technique uses strong magnetic fields and radio waves to generate images of the organs in the body? (1)
- A. X-Ray B. Ultrasound
C. CT Scan D. MRI

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8. Physics is considered the backbone of modern science and technology because its fundamental laws are applied in chemistry, biology, engineering, medicine, and communication technologies. Which of the following best explains why physics plays this central role? (1)
- A. Physics studies only matter and energy
 - B. Physics provides quantitative laws governing natural phenomena
 - C. Physics is older than other sciences
 - D. Physics replaces all other sciences
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9. In defense technology, ballistics is used to calculate the trajectory of a missile. If a missile is fired at an angle of 45° , which physics principle is primarily applied? (1)
- A. Archimedes' Principle
 - B. Projectile Motion
 - C. Pascal's Law
 - D. Bernoulli's Principle
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10. At the maximum height of a projectile launched at an angle, the velocity vector is: (1)
- A. Zero
 - B. Vertical
 - C. Horizontal
 - D. Tangential to the ground
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11. A centrifuge in a medical lab rotates at 3000 revolutions per minute (rpm). Calculate the approximate centripetal acceleration experienced by a sample at a radius of 0.10 m. (1)
- A. $10,000 \text{ m/s}^2$
 - B. $10,000,000 \text{ m/s}^2$
 - C. $1,000 \text{ m/s}^2$
 - D. $100,000 \text{ m/s}^2$
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12. You are designing a movie stunt. To make a motorcycle jump reach 4 times the previous maximum height ($4H$), how must you change the launch speed? (assume that the air resistance is neglected, launch angle remains constant) (1)
- A. Increase speed by 2 times
 - B. Keep speed the same
 - C. Increase speed by 4 times
 - D. Increase speed by 16 times
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13. You are designing a satellite orbit. To make the satellite take 8 times longer to complete one circle, you must expand the radius to: (1)
- A. 2 times the original radius
 - B. 64 times the original radius
 - C. 8 times the original radius
 - D. 4 times the original radius
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14. A ball is projected from the ground with an initial speed at an angle θ above the horizontal. Neglecting air resistance, the horizontal and vertical motions of the ball are independent because (1)
- A. Gravity acts only horizontally
 - B. Gravity affects only the vertical motion
 - C. No force acts on the projectile
 - D. Velocity remains constant
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15. A projectile's range is defined as its horizontal displacement when it returns to its original launch height. For two projectiles launched with the same initial speed, the ranges will be equal for these launch angles EXCEPT: (1)
- A. 15° and 75°
 - B. 20° and 70°
 - C. 40° and 50°
 - D. 40° and 60°
-
16. In analyzing the parabolic trajectory of a projectile launched over level ground with no air resistance, which of the following statements is categorically false? (1)
- A. The vertical component of velocity is zero at the maximum height.
 - B. The horizontal component of acceleration is constant and non-zero.
 - C. The time taken to reach maximum height is exactly half the total time of flight.
 - D. The speed at launch is equal to the speed just before impact.

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17. Two identical balls are launched simultaneously from the same height. Ball A is launched horizontally at speed u . Ball B is dropped from rest. Neglecting air resistance, which of the following is true about their motion just before they hit the ground? (1)
- A. Ball A has a greater vertical velocity component.
 - B. Ball B has a greater acceleration.
 - C. Both have the same vertical velocity component.
 - D. Ball A hits the ground first because it has an initial velocity.
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18. Consider a projectile launched with speed u at angle θ . Derive an expression for its maximum height H . Which derivation is correct? (1)
- A. Using $v^2 = u^2 + 2ay$, with $V = 0$, $U = u\sin\theta$, $a = -g$, gives $H = u^2\sin^2\theta/(2g)$
 - B. The maximum height is found when $X = u^2\sin 2\theta/(2g)$, leading to $H = u^2\sin^2\theta/(2g)$
 - C. Using the total energy, $H = u^2/(2g)$
 - D. The time to max height is $t = u/g$, leading to $H = u\sin\theta \cdot (u/g) = u^2\sin\theta/g$
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19. Which expression correctly combines the horizontal velocity (V_x) and time of flight (T) to find the Range (R)? (1)
- A. $R = V_x \cdot T$
 - B. $R = V_x/T$
 - C. $R = V_x + T$
 - D. $R = V_x - T$
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20. You are designing a mission to a planet of mass M . You need the satellite to orbit with a specific time Period T . Which formula derives the required orbital radius (r)? (1)
- A. $r = (GMT^2/4\pi^2)^{1/3}$
 - B. $r = (GM/T^2)^{1/3}$
 - C. $r = GMT/4\pi^2$
 - D. $r = (GMT^2)^{1/2}$
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21. The physical quantity that measures a body's resistance to changes in its rotational motion, serving as the rotational equivalent of mass, is called: (1)
- A. Torque
 - B. Angular Momentum
 - C. Moment of Inertia
 - D. Angular Velocity
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22. According to Kepler's First Law of Planetary Motion, the orbits of the planets are ellipses with the Sun located at: (1)
- A. The exact center of the ellipse
 - B. The surface of the planet
 - C. Point of maximum velocity
 - D. One of the two foci
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23. The equation for the trajectory of a projectile launched from the origin is $Y = x\tan\theta - gx^2/(2u^2\cos^2\theta)$. Deriving from this, what condition must be met for a projectile launched from ground level to pass through a specific point (X, Y)? (1)
- A. The point must satisfy the inequality $Y \leq X^2\tan^2\theta/(4H)$
 - B. The horizontal distance X must always be less than the maximum range $R_{\max}$
 - C. The vertical height Y must be zero for any real solution.
 - D. The launch angle must satisfy a quadratic equation in $\tan\theta$ derived from plugging (X, Y) into the trajectory equation.
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24. The gravitational force between two 1000 kg spherical masses whose centers are separated by 2.0 m is (1)
- A. 1.67×10^{-5} N
 - B. 3.34×10^{-5} N
 - C. 1.67×10^{-4} N
 - D. 3.34×10^{-4} N

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- 25.** A student claims that 'A projectile launched at 45° always achieves the maximum possible range regardless of air resistance.' Is this claim correct? (1)
- A. Yes, 45° is always optimal.
 - B. No, air resistance usually requires an angle slightly less than 45° .
 - C. No, the optimal angle is always 60° .
 - D. Yes, but only if the launch height is zero.
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- 26.** A satellite revolves around the Earth in a circular orbit. The centripetal force required for this motion is provided by (1)
- A. Atmospheric pressure
 - B. Earth's magnetic field
 - C. Gravitational force
 - D. Centrifugal force
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- 27.** For an object in uniform circular motion, the centripetal acceleration vector is characterized by being: (1)
- A. Constant in magnitude and direction.
 - B. Constant in magnitude, but always directed radially inward toward the center of the circle.
 - C. Increasing in magnitude as the object's speed increases.
 - D. Tangent to the path and opposite the direction of motion.
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- 28.** Kepler's Second Law is a direct consequence of the conservation of which physical quantity in the planet-Sun system, assuming no external torques? (1)
- A. Total Energy
 - B. Linear Momentum
 - C. Angular Momentum
 - D. Mass
-
- 29.** A ball is kicked with an initial velocity vector $\mathbf{v} = (30\mathbf{i} + 40\mathbf{j})$ m/s. What is the velocity of the ball at the highest point of its trajectory? (1)
- A. 0 m/s
 - B. 30 m/s
 - C. 40 m/s
 - D. 50 m/s
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- 30.** Which of the following graphs best represents the relationship between the gravitational force (F) exerted by a planet and the distance (r) from its center, for $r > R_{\text{planet}}$? (1)
- A. A straight line with positive slope.
 - B. A straight line with negative slope.
 - C. A hyperbola (curve dropping as $1/r$).
 - D. An inverse square curve (curve dropping as $1/r^2$).
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- 31.** A ship floats higher in saltwater than in freshwater. This is because: (1)
- A. The buoyant force is greater in saltwater due to its higher viscosity.
 - B. Saltwater is denser, so a smaller volume needs to be displaced to equal the ship's weight.
 - C. The gravitational acceleration is lower over the ocean.
 - D. The hull of the ship is specially designed to repel salt molecules.
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- 32.** In a horizontal pipe with a constriction (a venturi), the fluid velocity increases in the narrow section. According to Bernoulli's principle, this increase in velocity is accompanied by: (1)
- A. A decrease in static pressure.
 - B. An increase in static pressure.
 - C. No change in static pressure if the flow is ideal.
 - D. An increase in the fluid's density.

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- 33.** Why is the dam wall thicker at the bottom than at the top? (1)
- A. Because pressure in a static fluid increases linearly with depth.
 - B. To support the weight of the wall itself.
 - C. To prevent the water from freezing at the bottom.
 - D. Because the speed of water flow is higher at the bottom.
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- 34.** Two discs of the same material and thickness have radii 0.2 m and 0.6 m. Their moments of inertia about their axes will be in the ratio of (1)
- A. 1 : 81
 - B. 1 : 27
 - C. 1 : 9
 - D. 1 : 3
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- 35.** If the average distance from the Sun (semi-major axis) of an asteroid is 4 times that of Earth, what is its orbital period according to Kepler's Third Law? (Earth's period = 1 year) (1)
- A. 4 years
 - B. 8 years
 - C. 16 years
 - D. 64 years
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- 36.** At what altitude above Earth's surface would the acceleration due to gravity be half its value at the surface? (Assume Earth's radius is R). (1)
- A. R
 - B. R/2
 - C. $R\sqrt{2}$
 - D. $R(\sqrt{2} - 1)$
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- 37.** The maximum height reached by a projectile depends mainly on (1)
- A. Horizontal component of velocity
 - B. Distance and angle
 - C. Vertical component of velocity
 - D. Mass of projectile
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- 38.** If the distance between two masses is doubled, the gravitational force between them becomes (1)
- A. Twice as large
 - B. Half as large
 - C. One-fourth as large
 - D. Four times larger
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- 39.** Newton's law of gravitation applies to (1)
- A. Only planets
 - B. Only celestial bodies
 - C. All masses
 - D. Charged particles only
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- 40.** Three identical masses are placed at the corners of an equilateral triangle of side L. What is the magnitude of the net gravitational force on any one of the masses due to the other two? (1)
- A. $\sqrt{3} \times Gm^2/L^2$
 - B. Gm^2/L^2
 - C. $\sqrt{3}Gm^2/4L^2$
 - D. $2Gm^2/L^2$
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- 41.** You are designing a fire hose nozzle. To make the water shoot out 4 times faster than it does now, how must you change the cross-sectional area of the opening? (1)
- A. Make the area 4 times larger.
 - B. Make the area 2 times larger.
 - C. Make the area 1/4 of the original size.
 - D. Make the area 1/2 of the original size.
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- 42.** A town's water tower is an open tank located 30 m above the ground. The water in the tank is 5 m deep. What is the water pressure at a neighborhood that is 20 m above the ground? (1)
- A. 100 kPa
 - B. 150 kPa
 - C. 200 kPa
 - D. 250 kPa

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- 43.** In a hydraulic lift, a small piston has an area of 2 m^2 and a large piston has an area of 10 m^2 . If a force of 20 N is applied to the small piston, what force is exerted by the large piston? (1)
- A. 4 N B. 20 N
C. 40 N D. 100 N
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- 44.** Gauge pressure is equal to: (1)
- A. Absolute pressure + Atmospheric pressure B. Absolute pressure – Atmospheric pressure
C. Atmospheric pressure – Absolute pressure D. Absolute pressure $\times$ Atmospheric pressure
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- 45.** In a horizontal pipe, the pressure at point A (slow water) is P_a and point B (fast water) is P_b . Which is true? (1)
- A. $P_a > P_b$ B. $P_a < P_b$
C. $P_a = P_b$ D. $P_a = 0$
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- 46.** A U-tube manometer contains mercury. One end is open to the atmosphere, and the other is connected to a gas tank. The mercury level in the arm connected to the gas is 15 cm lower than in the open arm. What is the gauge pressure of the gas? (1)
- A. 20.4 kPa B. 10.2 kPa
C. 2040 Pa D. 1020 Pa
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- 47.** A submarine is designed to withstand an absolute pressure of $8 \times 10^6 \text{ Pa}$. At what approximate depth in seawater would it reach its crush depth? (1)
- A. 700 m B. 770 m
C. 800 m D. 880 m
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- 48.** A solid object weighs 50 N in air and 30 N when fully immersed in water. What is the density of the object? (1)
- A. 1250 kg/m^3 B. 2000 kg/m^3
C. 2500 kg/m^3 D. 5000 kg/m^3
-
- 49.** An airplane wing is designed so that air flows faster over the top surface than the bottom. Using Bernoulli's principle, the primary reason this design generates lift is that: (1)
- A. The higher velocity above creates a region of lower pressure relative to the higher pressure below the wing.
B. The air molecules are deflected downward, and by Newton's third law, the wing is pushed upward.
C. The stagnation point moves to the lower surface, increasing pressure there.
D. The difference in velocities creates a net upward drag force.
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- 50.** Water flows steadily through a horizontal pipe that narrows from a diameter of 10 cm to a diameter of 5 cm . If the speed in the wide section is 2 m/s , what is the speed in the narrow section? (1)
- A. 4 m/s B. 6 m/s
C. 8 m/s D. 10 m/s
-
- 51.** Two immiscible liquids of densities $\rho_1 > \rho_2$ in a U-tube. At equilibrium (1)
- A. The heavier liquid rises higher B. The lighter liquid rises higher
C. Zero pressure difference D. Same level always

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- 52.** Derive the expression for the total pressure at a depth (h) in a fluid of density (ρ) that is open to the atmosphere with pressure (P_{atm}). Which derivation is correct? (1)
- A. The pressure is the sum of atmospheric pressure and the pressure from the fluid column: $P = P_{\text{atm}} + \rho gh$
 - B. The pressure is due to the fluid column only: $P = \rho gh$.
 - C. The pressure decreases with depth: $P = P_{\text{atm}} - \rho gh$.
 - D. Pressure is constant in a static fluid, so $P = P_{\text{atm}}$ at all depths.
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- 53.** A cylindrical water tower of height H is filled to the brim. A small leak develops at the very bottom. Derive the expression for the speed of the water exiting the leak (Torricelli's theorem). The derivation uses Bernoulli's equation between the free surface (point 1) and the leak (point 2), assuming both are open to the atmosphere, leading to: (1)
- A. $v = \sqrt{2gH}$
 - B. $v = gH$
 - C. $v = \sqrt{H}$
 - D. $v = 2gH$
-
- 54.** Why does atmospheric pressure decrease as altitude increases? (1)
- A. Air becomes colder.
 - B. The gravitational pull increases.
 - C. The weight of the air column above decreases and air density drops.
 - D. The wind speed increases at high altitudes.

— END OF EXAMINATION —